NAME:KRISHNAPRIYA P

REG NO:192511279

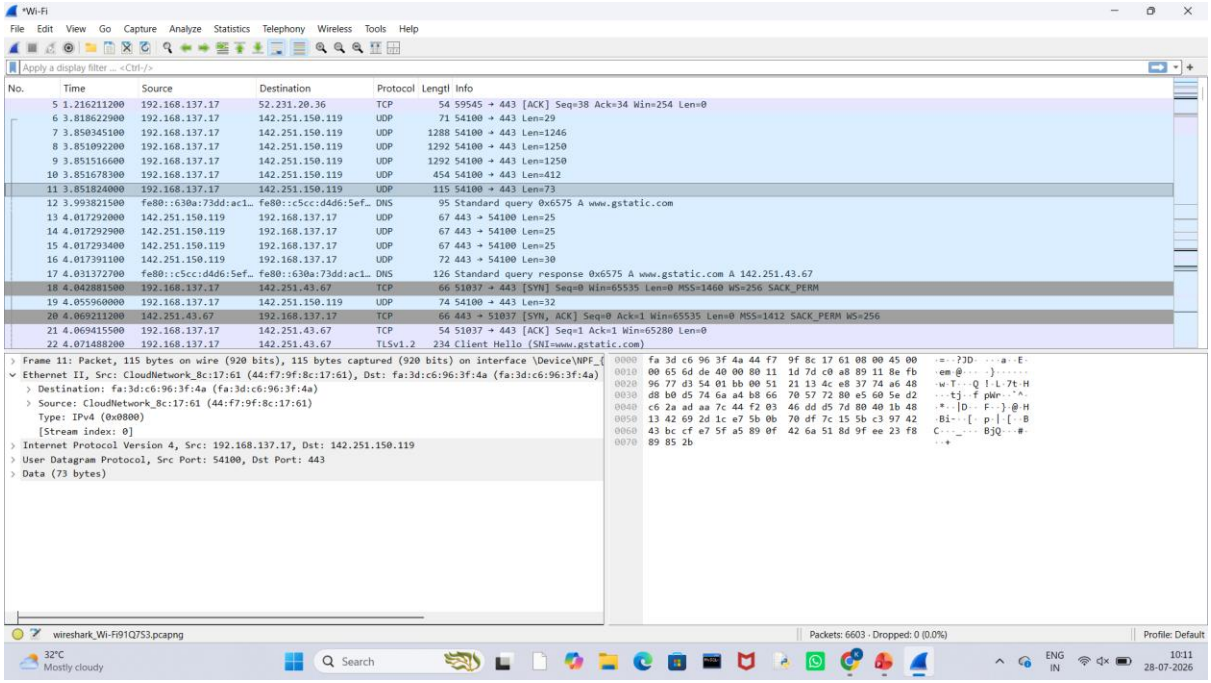
COURSE CODE:CSA0712

COURSE NAME:COMPUTER NETWORKS

CO2-AT1- Analytical Problem Solving

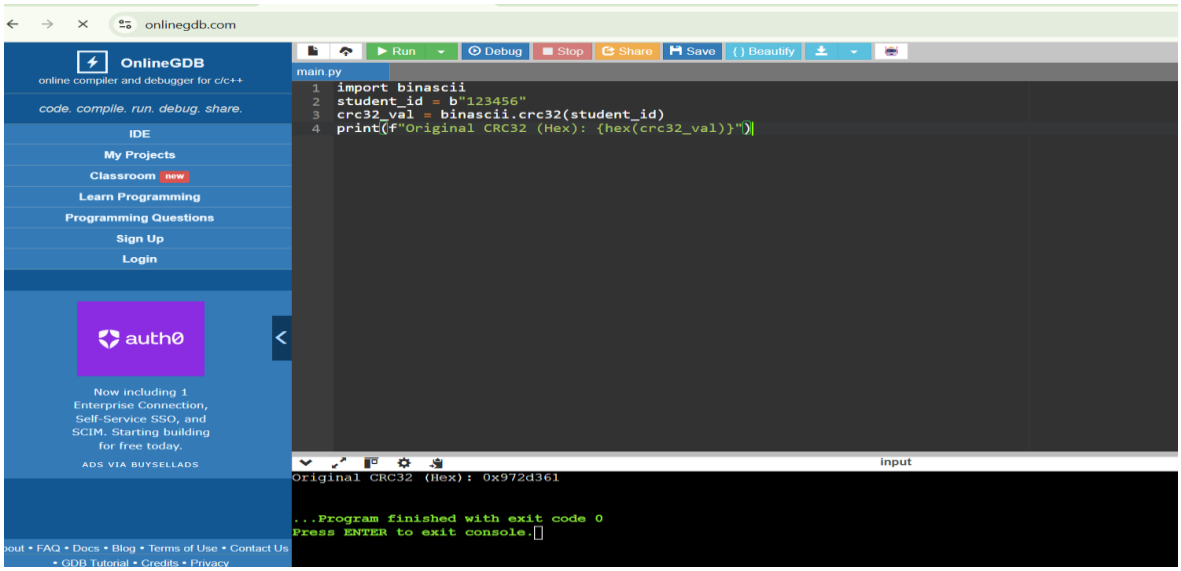
Q1. Live Frame Checksum Verification

- Tool: Wireshark (Local or Cloud-hosted like CloudShark)
- Task: Capture or open a PCAP file containing HTTP traffic. Locate an Ethernet frame. Find the Frame Checksum Sequence (FCS) or Header Checksum field. Change Wireshark's preferences to enforce validation. Explain what happens visually in the GUI if you manually flip a bit in a hex editor before opening it.



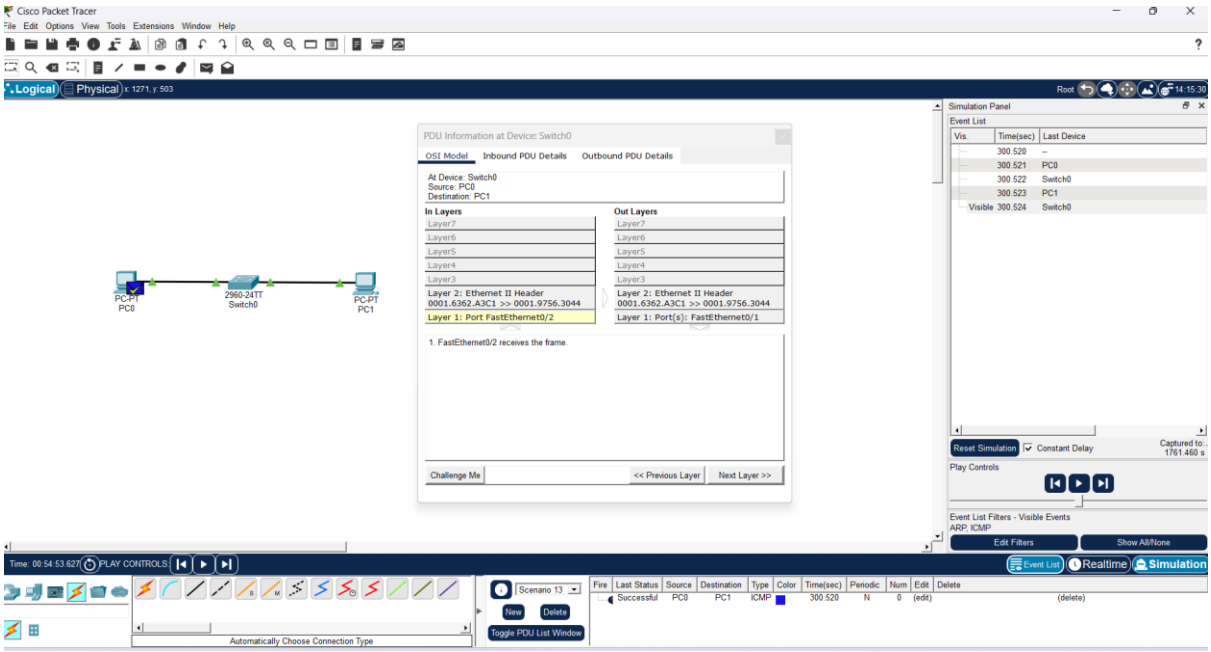
Q5. Python-Based CRC-32 Validation

- Tool: Online Python IDE (Repl.it or Jupyter Notebook Online)
- Task: Use the binascii library in Python to calculate the CRC-32 checksum of your student ID string. Modify one character in your ID and run the script again. Record the absolute numerical difference between the two 32-bit outputs to demonstrate the avalanche effect.



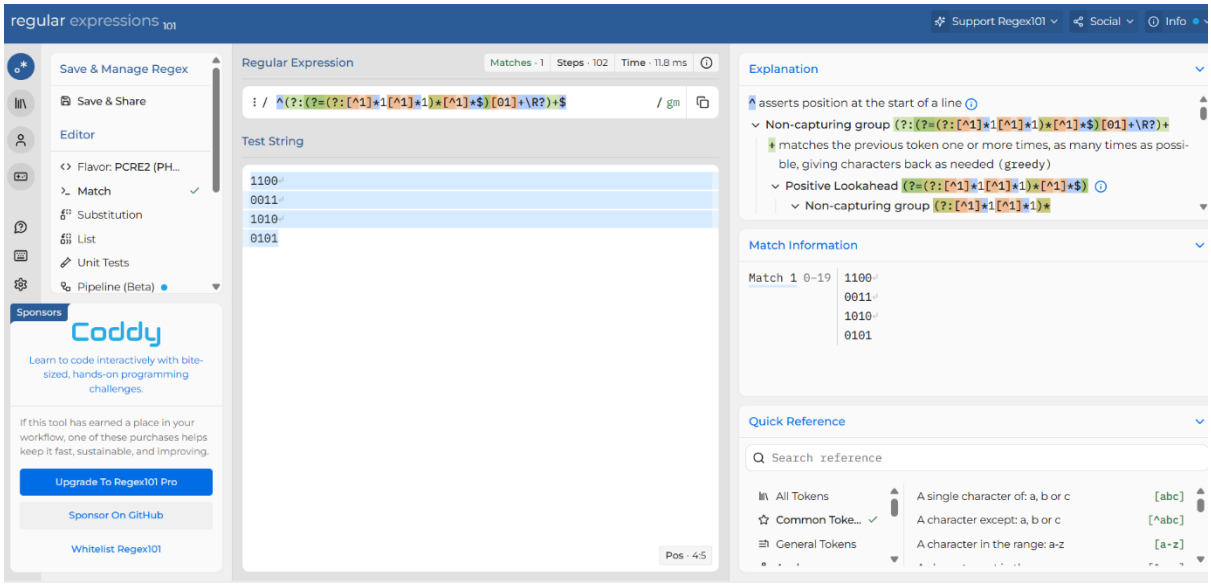
Q7. Packet Tracer FCS Drop Test

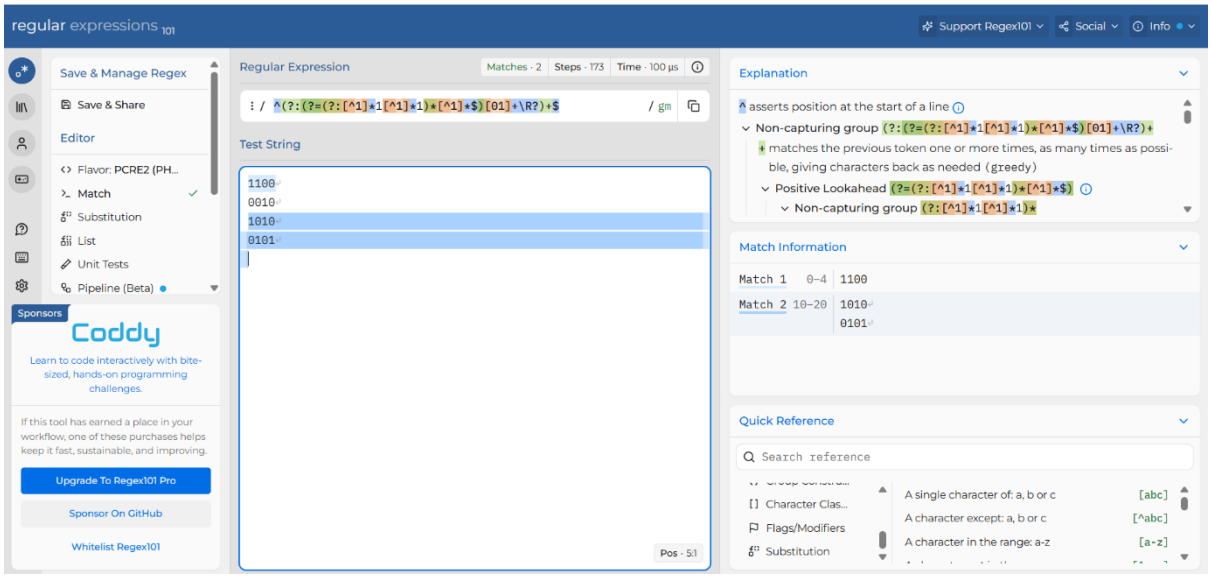
- Tool: Cisco Packet Tracer (or NetSim)
- Task: Set up a simple 2-PC network. Go to "Simulation Mode". Send a ping packet. Click the packet envelope at the switch to inspect the Inbound/Outbound PDU details. Locate the Layer 2 FCS. Take a snippet showing the Switch hardware validating this field.



Q6. Regex Error Pattern Matcher

- Tool: Regex101.com
- Task: Write a regular expression that validates if a received string matches a valid 2D even parity block layout. The regex must flag a syntax error if any row contains an odd number of 1s. Paste your functional regex link.


regular expressions 101
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Save & Share
Editor
Flavor: PCRE2 (PH...
Match
Substitution
List
Unit Tests
Pipeline (Beta)
Regular Expression
Matches · 1
Steps · 102
Time · 11.8 ms
/ ^(?:(?=(?:[^1]*1[^1]*1)*[^1]*$)[01]+\R?)+$ / gm
Test String
1100
0011
1010
0101
Pos · 4:5
Explanation
^ asserts position at the start of a line
Non-capturing group (?:(?=(?:[^1]*1[^1]*1)*[^1]*$)[01]+\R?)+
+ matches the previous token one or more times, as many times as possible, giving characters back as needed (greedy)
Positive Lookahead (?=(?:[^1]*1[^1]*1)*[^1]*$)
Non-capturing group (?:[^1]*1[^1]*1)*
Match Information
Match 1 0-19
1100
0011
1010
0101
Quick Reference
Search reference
All Tokens
Common Toke...
General Tokens
A single character of: a, b or c [abc]
A character except: a, b or c [^abc]
A character in the range: a-z [a-z]
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Match
Substitution
List
Unit Tests
Pipeline (Beta)
Regular Expression
Matches · 2
Steps · 173
Time · 100 µs
/ ^(?:(?=(?:[^1]*1[^1]*1)*[^1]*$)[01]+\R?)+$ / gm
Test String
1100
0010
1010
0101
Pos · 5:1
Explanation
^ asserts position at the start of a line
Non-capturing group (?:(?=(?:[^1]*1[^1]*1)*[^1]*$)[01]+\R?)+
+ matches the previous token one or more times, as many times as possible, giving characters back as needed (greedy)
Positive Lookahead (?=(?:[^1]*1[^1]*1)*[^1]*$)
Non-capturing group (?:[^1]*1[^1]*1)*
Match Information
Match 1 0-4 1100
Match 2 10-20 1010
0101
Quick Reference
Search reference
Character Clas...
Flags/Modifiers
Substitution
A single character of: a, b or c [abc]
A character except: a, b or c [^abc]
A character in the range: a-z [a-z]
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